

Pneumonia Detection from Chest X-Ray images using DenseNet

Abstract

In the world, pneumonia is considered a major health problem, where experienced radiologists are not easily available in some regions, so detecting the disease early will help the patient's recovery faster. currently the doctor examines the chest X-ray manually, which is time consuming and can make errors. To address this issue, the study proposes an automated pneumonia detection system based on deep learning. DenseNet is used because it can extract the important features effectively and allows the features to transfer between the layers. The model is trained and evaluated using a widely recognized chest X-ray dataset. While comparing it with earlier models like ResNet and Vision Transformer, its performance is high, and it achieves high accuracy. Grad-CAM heatmaps are used so that doctors can see where the model is focusing on the lungs. A structured medical report is generated for each case, which consists of the original chest X-ray image, the corresponding heatmap, and the predicted confidence score. The proposed system provides a reliable, explainable, and user friendly diagnostic tool to assist clinicians in real world healthcare environment.

Keywords: Chest X-Ray, Deep learning, DenseNet, ResNet, Vision Transformer, Grad-CAM, Pneumonia

1.Introduction

Recently, research shows that using deep learning, it is possible to detect pneumonia from chest X-ray images. In early work, many people used Convolutional Neural Networks (CNN) models like VGG, ResNet, and DenseNet to automatically extract important features from X-ray images to detect pneumonia which helped in diagnosing diseases [2], [5], [10]. While looking at other research papers model performance technique was improved by transfer learning, combining multiple models, and tuning the training process, all these things were applied on well known datasets such as NIH Chest X-ray14 and the Guangzhou pediatric X-ray dataset [2], [6], [9], [10]. Models automatically look only at the important parts of the X-ray instead of looking at the whole image. Focus on the important area, and enhance the area before sending the image to the model, so the model learns better. Combining two or more models (CNN) produces good performance [8], [15]. To understand where the model made the prediction, a technique where used, which is Grad-Cam, creates heatmaps showing which parts of the lungs the model where looking at and made the prediction [10], [11]. Some research even created an interface where the user uploads an X-ray image and sees the prediction made by the model interactively [12]. In numerous reviews, the potential of deep learning for pneumonia screening has been documented, while acknowledging critical gaps including dataset imbalance, domain shift, label noise, insufficient real world clinical validation, and a lack of diverse clinical applications [13], [3], [14].

Despite this progress, several gaps still exist. Many such systems perform very well on a single public dataset with limited or no cross-domain evaluation to establish real-world reliability or confidence in actual deployment across different hospitals and varied imaging contexts [2], [6], [9], [13]. Interpretability in most works is confined to saliency maps without structured clinical context or transparent justifications that align with radiologist decision making processes, thus limiting clinical trust and explainability [10], [11], [15]. Lastly, though a few exist, most prior works do not present standardized clinician-oriented reporting that incorporates original chest X-ray images, explanation heatmaps, decision certainty, and patient metadata into one consolidated diagnostic output [12], [6]. These gaps call for the development of models with high accuracy across diverse datasets and with effective clinically relevant explanation of their decisions.

In this work, an explainable DenseNet based framework for automatic pneumonia detection is proposed. The system leverages DenseNet for efficient gradient flow and feature propagation for enhancing the representation capability of subtle radiographic patterns. The system uses Grad-CAM, which highlights the region where the model focus and made its prediction, which helps the doctors to understand the results. For each prediction, a medical report as a PDF will be generated, which includes the original X-ray image, heatmap, highlighted important regions, predicted class, confidence score, and optional patient information. Combining good model performance, clear explanations, and detailed reporting allows us to move from research into practical hospital use.

2. Literature Review

Ref	Title	Method	Accuracy	Contribution
[1]	Detection of Pneumonia Using Convolutional Neural Networks and Deep Learning	CNN	97%	High pediatric pneumonia accuracy using CNN layers
[2]	Pneumonia Detection Using Convolutional Neural Networks	VGG / ResNet	92%	TL advantages in pneumonia classification
[3]	Pneumonia Detection Using Deep Learning Approaches	CNN	94%	Systematic evaluation of deep models for pneumonia
[4]	Pneumonia Detection: An Efficient Approach Using Deep Learning	CNN + augmentation	96%	Improved accuracy through augmentation + preprocessing
[5]	A DL based Model for Detection of Pneumonia using VGG-16 & NN	VGG16 + Dense layers	95%	Transfer learning with VGG-16 showing high performance
[6]	Efficient Pneumonia Detection in Chest X-ray Images Using Deep TL	ResNet/MobileNet/Dense Net	98.4%	Ensemble models boosting classification precision
[7]	Deep Learning for Automatic Pneumonia Detection	CNN	93%	automated pneumonia feature extraction
[8]	Detection & Classification of Lung Diseases for Pneumonia & COVID-19	CNN + ML hybrid	93%	Combined CNN + RNN for lung disease classification
[9]	Detecting Pneumonia in Chest X-Rays with Supervised Learning	DenseNet + imbalance handling	AUC > 0.90	Studied class imbalance + patientsplit training
[10]	CheXNet: Radiologist-Level Pneumonia Detection on CXRs	DenseNet-121 + Grad-CAM	Radiologistlevel F1	First radiologist-level deep learning model
[11]	Pneumonia Detection Using Deep Learning	CNN	91%	baseline CNN performance for pneumonia
[12]	Pneumonia Detection Using Deep Learning	CNN + data augmentation	92%	improvement via augmentation strategies
[13]	Review on Pneumonia Image Detection: A ML Approach	Review Study	-	imbalance & domain shift
[14]	Pneumonia Detection Using DL Based on CNN	CNN + TL	94%	Strong CNN-based TL performance analysis
[15]	Pneumonia Classification Using DL from CXRs During COVID-19	CNN + TL	96%	Multi class classification including COVID pneumonia

Table 1: Comparison of various Pneumonia detection techniques

Deep learning is widely used for detecting pneumonia using chest X-ray images because CNNs work well in medical images. The early studies show that CNNs can work very well on important features for pneumonia [1], [2]. Some research applied transfer learning using models such as VGG 16, ResNet, and achieved good results when working with small datasets [2], [5]. Deep learning models have improved by using data augmentation, optimized hyperparameters, loss functions to handle class imbalance, and improving the model's ability to perform well on new chest X-ray images [3], [4], [6]. Using multiple models together or combining different types of deep learning architectures produces good output [6], [8]. At the same time, research shows that Densenet works well on pneumonia detection and produces good performance while comparing to other models. The deep learning models can handle large datasets, which will be effective for real clinical interpretation [10].

Many studies show that Grad-CAM highlights the lung regions in chest X-rays that show where the model made its prediction [10], [11]. Researchers are also creating graphical user interfaces that make the doctor to understand easily [12]. Some researchers have pointed out several challenges, like uneven class distribution, differences between training data and real hospital data, and errors or noise in the labels [3], [13], [14].

There is some progress, but there are some limitations. Many models are trained and tested on the same dataset, which affects the performance [2], [6], [9], [13]. A Grad-CAM is used to show the virtual heatmap, which shows where the model made its prediction, not confirmation [10], [11], [15]. The system also provides an easy interface, so every user can use it easily. The model generates complete medical reports for healthcare. These reports include chest X-ray images, heatmaps, and confidence scores. which can be used in the real world [6], [12]. Future work should not only focus on high accuracy but also on being easy to understand, transparent in their decisions, and able to fit into current hospital systems and workflows.

3.Dataset

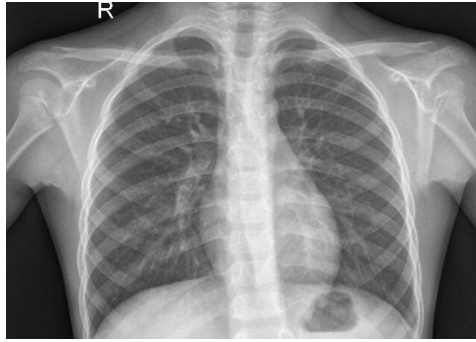
The chest X-ray dataset is obtained from Kaggle, which contains two classes normal and pneumonia. A total of 7,750 images are used, with an equal number of normal and pneumonia images, where present, by keeping the dataset balanced helps the model learn both classes fairly and to produce a robust model and also reduces bias towards one class. All images were pre-processed resizing and normalizing them to the same dimension pixel Intensity value. to enhance the model Generalization capability, various data enhancements Random rotation, horizontal flipping and Zooming was implemented during training. The dataset was split into three subgroups 70% for training, 20% for validation, and 10% for testing. This structured partitioning allowed the model to Learn, tune, and evaluate hyperparameters effectively Performance on unseen data while preventing overfitting.

3.1 Normal Class

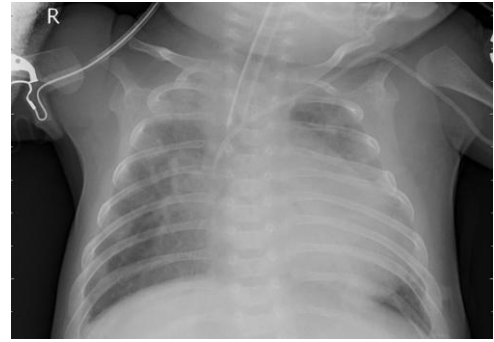
The normal class contains 3,875 X-ray images representing healthy lungs without any signs of infection or Abnormal opacity. These X-rays usually show clear lungs fields, visible rib structures, and specific heart and Outline of diaphragm. They serve as a control group Training and testing the model. As shown in (figure 1).

3.2 Pneumonia Class

There are also 3,875 images in the Pneumonia category, Illustration of lungs infected with pneumonia. These x-rays Displays patterns such as opacity areas, contours, and Inflammation, usually caused by bacteria or viral Infection. The infected lung area appears denser or whiter More than normal due to accumulation of fluid in the alveoli. As shown in (figure 1).



Normal



Pneumonia



Normal



Pneumonia

figure 1: sample X-ray images from both classes Normal and Pneumonia. The normal X-ray shows clear and well defined lung regions, whereas the pneumonia X-ray displays visible cloudy regions and dense opacities, indicating infection.

4. Methodology

The proposed system uses a deep learning based approach to automatically detect pneumonia from chest X-ray images. The system not only predicts pneumonia but also explains the result and presents it clearly for doctors. The entire model works in four stages. First, the data is preprocessed. Next, the model is trained using the data. After training the model produce the heatmap with the help of Grad-CAM, which helps to identify which area does the model made its prediction. Next, the model produces a medical report as a PDF, so the entire detail regarding the prediction can be viewed.

s4.1 Data Preprocessing

Before training, all the chest X-ray images will be preprocessed, which is performed by resizing all the images to the same dimensions and their pixel value are scaled between 0-1, which allows the training to be more stable and faster. Data augmentation technique is applied so the model will perform on new unseen data. These include small rotations, flipping, zooming, and contrast changes. This allows the model to learn on different variations, which is required for the real world. This will be done without losing the important data.

4.2 Model Architecture

The used framework is DenseNet, which helps the information pass through each network layer and reduces the problem during training, and performs well on medical images. The DenseNet allows each and every layer to share the output, which allows the model to reuse learned features and keeps the network smaller and efficient.

The output of the l^{th} layer is defined as:

$$x = \text{HL}([x_0, x_1, x_2, \dots, x_{l-1}])$$

In this expression $\text{HL}(\cdot)$ refers to batch normalization, ReLU activation, convolution operation, and $[\cdot]$ indicates feature map concatenation. This allows each layer can use information from the previous layer, which gives better gradient flow and improved feature learning.

To maintain computational efficiency while enabling deep feature extraction, the network employs transition layers between dense blocks that perform 1×1 convolution followed by 2×2 average pooling. After the final dense block, the architecture includes:

- Global Average Pooling layer
- Fully Connected dense layer
- Softmax output layer for binary classification (Normal vs Pneumonia)

This configuration enhances the network's ability to learn subtle structural differences in lung radiographs, particularly consolidations, infiltrations, and opacity patterns indicative of pneumonia, while ensuring stable training and efficient gradient flow. [\(figure 2\)](#) illustrates the DenseNet 121 architecture.

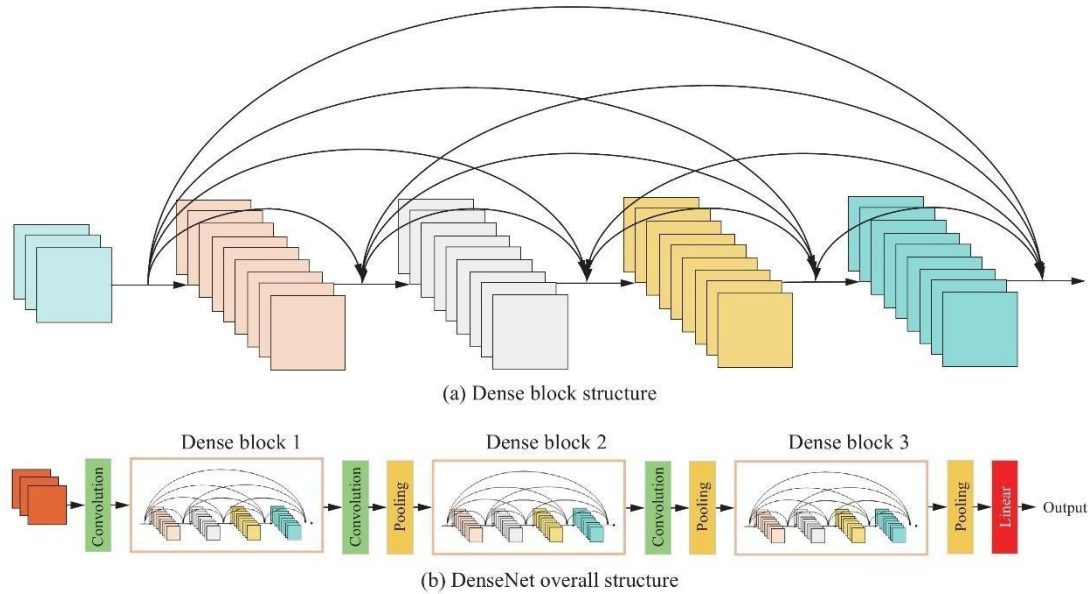


figure 2: DenseNet architecture

4.3 Training Configuration

The model was trained using a cross entropy loss function, and the Adam optimizer helped the model learn efficiently. Using gradient descent, small sets of data are processed to improve predictions during training and speed up the final results. Early stopping and learning rate determination were implemented to prevent overfitting and ensure optimal generalization. The validation set was used to adjust the model settings and choose the best performing model during training.

4.4 Explainability Module

To provide interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) changed into integrated into the gadget. Grad-CAM highlights discriminative areas of the lung that influence the version's selection, producing heatmaps overlaid at the unique X-ray. The model draws a circle around the area where the model confirms which area is affected. This makes the results easier and helps the doctors to understand where the model made its prediction, as shown in [\(figure 3\)](#).

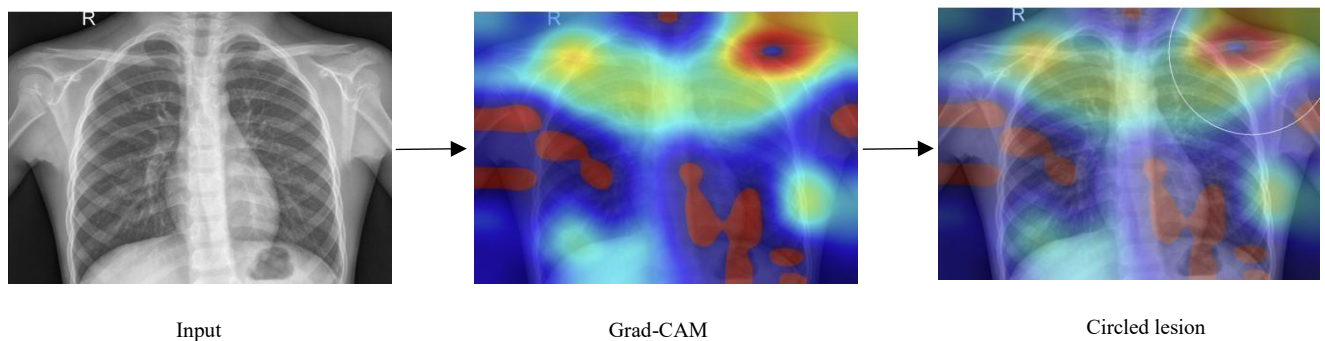


figure 3: Explainability Module

4.5 Diagnostic Report Generation

For each prediction, the model will generate a medical style PDF report as that includes:

- Original chest X-ray
- Grad-CAM heatmap overlay
- Highlighted lesion region
- Classification result (Normal/Pneumonia)
- Confidence score percentage
- Optional patient information fields (name, age, gender)

The prediction result will be in a report format that helps to understand easily, and also to be used in a hospital workflow as shown in [\(figure 4\)](#).

Chest X-ray Pneumonia — Case Report

Generated: 2025-11-03 15:34:25

Patient Name: ram

Age: 19

Sex: M

Case File: IM-0001-0001.jpeg

Model: DenseNet-121 + Grad-CAM Explainability

Prediction: NORMAL | $P(\text{pneumonia}) = 0.172$

Risk Level: Low

Severity (CAM-area proxy): Moderate | area $\approx 15.0\%$

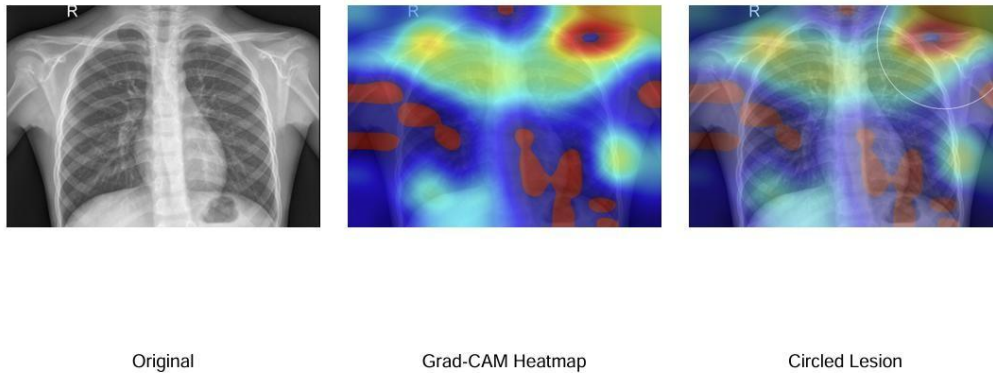


figure 4: Report made by the model

4.6 System Workflow

The entire pipeline of the model:

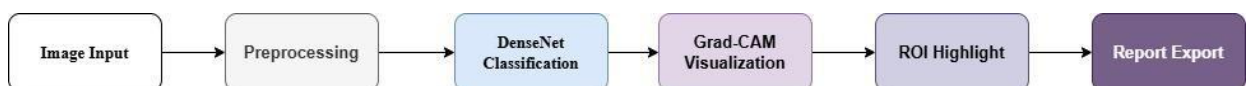


figure 5: Workflow

Stage 1: Image Input

The process begins with a chest X-ray, where the user provides their X-ray, which will be processed to the model.

Stage 2: Preprocessing

Before classification, the input chest X-ray image will be preprocessed to improve the quality of the image. where the preprocessing consists of resizing the image, removing noise, and normalizing, this ensures the perfect image for deep learning for the better output.

Stage 3: DenseNet Classification

The preprocessed images move to the Densenet, where the images are classified as either pneumonia or normal.

Stage 4: Grad-CAM Visualization

The Gard-CAM is used to generate a heatmap that highlights the lung region that contributes to the classification decision.

Stage 5: ROI Highlight

Regions of interest (ROI) are identified and visually marked on the chest X-ray. This helps to locate easily the suspected pneumonia-affected areas easily.

Stage 6: Report Export

Finally, a structured report includes the original X-ray image, highlighted regions, Grad-CAM heatmaps, and the model's confidence score, making the results easy to understand and clinically useful. The report is generated.

5. Results

The model was evaluated using standard classification performance metrics, including accuracy, precision, recall, F1-score, and the confusion matrix. This illustrates how the model performs on the two classes normal and pneumonia, based on chest X-ray images.

As shown in [\(table 2\)](#), the model achieved an overall accuracy of 92% which shows that it performs well on both classes. In the normal class, the model obtained a precision of 0.9426, a recall of 0.8491, and an F1-score of 0.8934 over 232 samples. The model is more accurate on the normal class and few false positives.

The pneumonia class, the model achieved a higher recall of 0.9692, along with a precision of 0.9153 and an F1-score of 0.9415, evaluated on 390 samples. With the high recall, the model can correctly identify the pneumonia cases.

Overall, the model can differentiate between normal and pneumonia using chest X-ray images, and also with balanced performance across both classes.

Class	Accuracy	Precision	Recall	F1-Score	Support
NORMAL	84.91%	0.9426	0.8491	0.8934	232
PNEUMONIA	96.92%	0.9153	0.9692	0.9415	390
Overall model	92.44%	0.9254	-	-	622

Table 2: Model Performance

5.2 Confusion Matrix

The model can easily identify both pneumonia and normal cases. As shown in [\(figure 6\)](#), there are a small number of pneumonia cases incorrectly classified as normal (12 false negatives), which reduces the chance of missing pneumonia diagnoses.

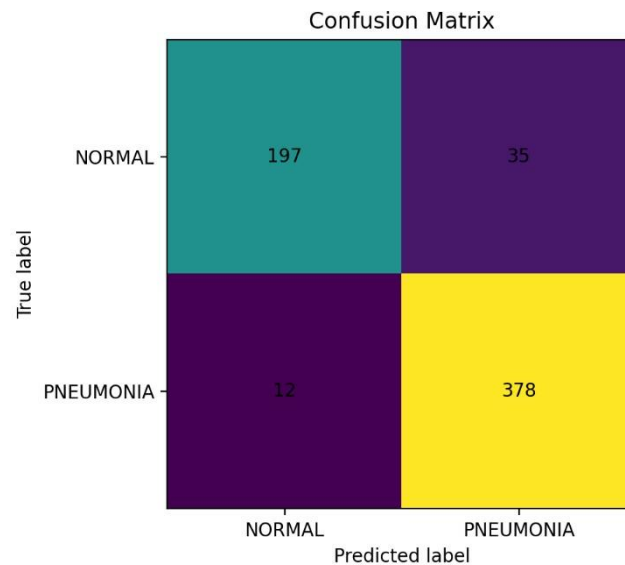


figure 6: Confusion Matrix

5.3 ROC-AUC Analysis

The ROC curve shows that the model is very good at distinguishing between pneumonia and normal chest X-ray images. An AUC value of 0.96, as shown in [\(figure 7\)](#), tells the difference between the two classes even when the decision threshold is changed.

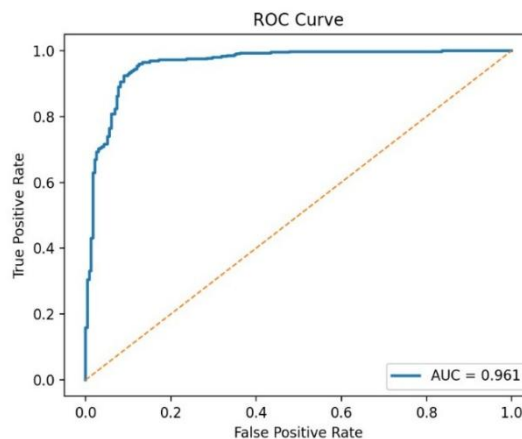


figure 7: ROC Curve

5.4 Visualization of Predictions

(Figure 8) illustrates represent examples of model predictions, showcasing:

- Correctly classified normal and pneumonia cases
- Activation regions over lung fields
- Visible areas of opacity and consolidation in pneumonia positive samples

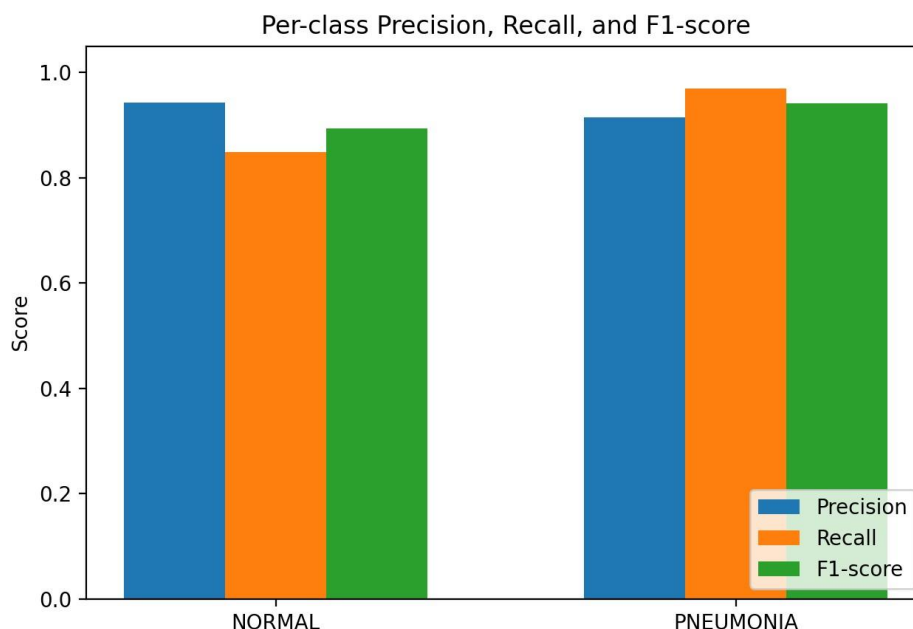


figure 8: Per-Class Prediction

6. Conclusion and Future Work

This study is about a deep learning system based on DenseNet using chest X-ray images to detect pneumonia. The system includes Grad-CAM to explain the model's decisions and generates structured medical reports as a PDF. The overall model has a good performance with an accuracy of 92% and a pneumonia detection rate of 96%. GradCAM and ROI highlight the affected part of the lungs, which will be helpful for the doctors to understand visually. In addition to that, the reporting system combines X-ray images, heatmaps, confidence scores, and patient information into a single report.

The model performs well it still needs further improvement before it can be fully used in real hospital settings. In the future model will be ill be extended to detect different types of lung diseases, not just pneumonia, and the plan is to develop a medical AI chatbot that can generate detailed radiology reports, suggest next steps, and provide personalized health guidance. This would allow not just detecting the disease and also a medical assistant as an AI that supports patients and doctor with clear information.

Overall, work is not about building technology and also on making healthcare information clearer and easier for people to understand. In future improvements, the model can be a complete health assistant that helps detect diseases at an early stage and supports patients.

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